



Silent Witness

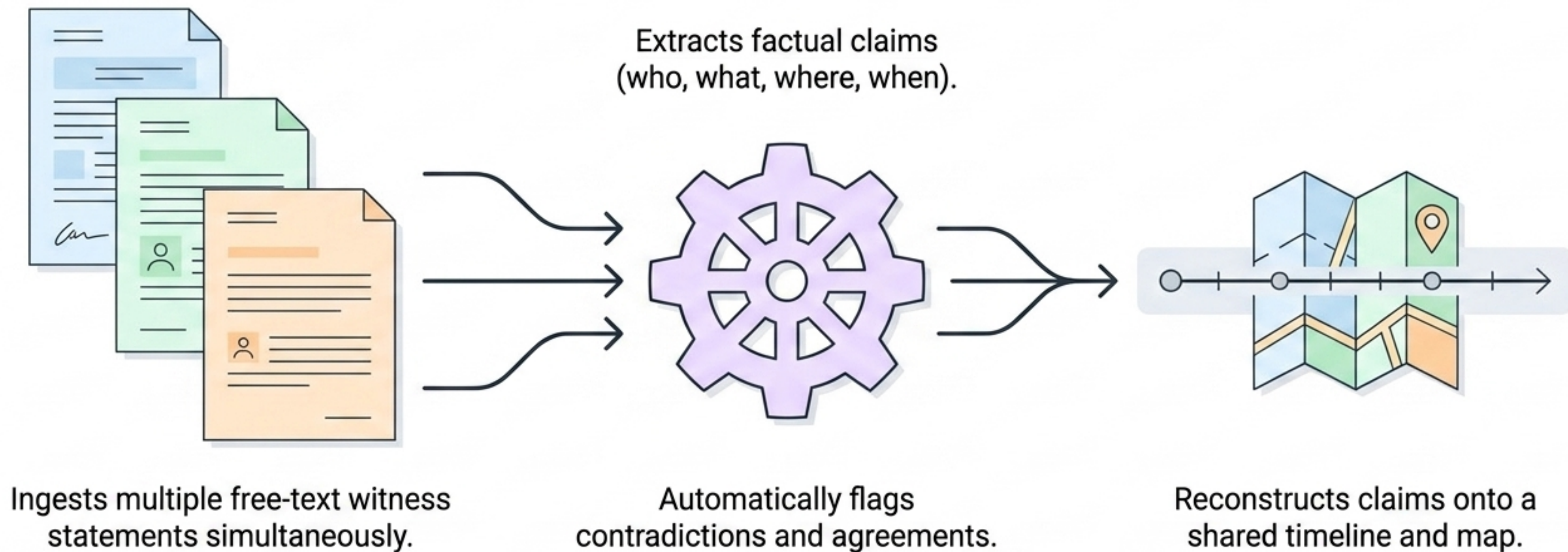
Testimony Consistency Reconstructor

The Problem: Human Accounts Diverge

Human memory is fallible and subjective, leading to inconsistent and conflicting accounts from multiple witnesses of the same event. Reconciling these discrepancies and identifying factual commonalities is currently a laborious, manual, and error-prone process that cannot easily scale with the volume of testimony or complexity of the event.

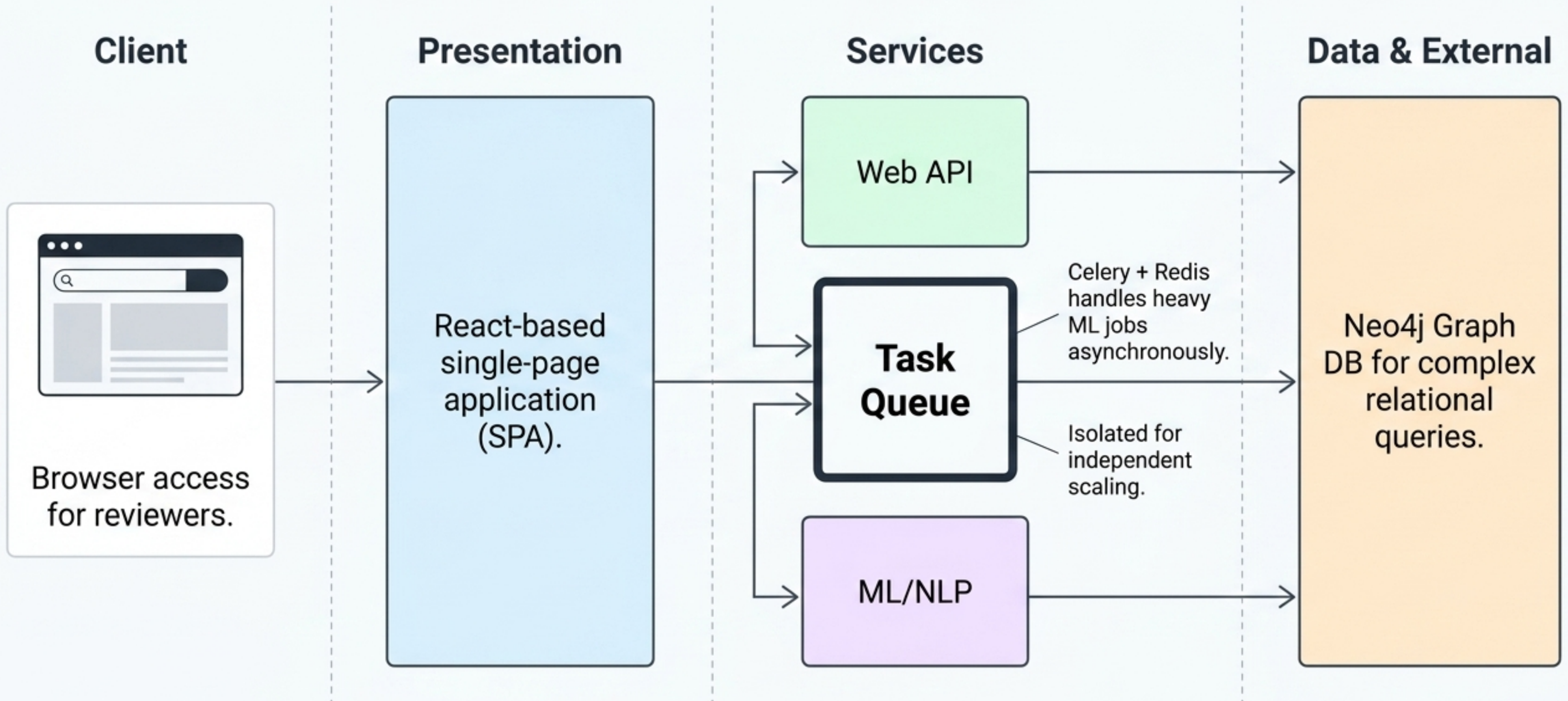
The current workflow relies on disconnected, analog, and fragmented digital tools like highlighters, physical whiteboards, and complex spreadsheets. This outdated approach severely bottlenecks the process, dramatically increasing the time-to-insight for journalists, investigators, and legal teams trying to construct a coherent narrative.

The Solution: Automated Extraction & Alignment



Strictly Non-Adjudicative: Evaluates consistency, not credibility.

System Architecture: Decoupled & Asynchronous



Technology Stack

Presentation

React

Tailwind CSS

Leaflet

vis-timeline

API / Backend

FastAPI / Express

JWT Auth

ML / NLP

spaCy

Hugging Face

sentence-transformers

NLI (roberta)

Data

Neo4j

PostgreSQL

Redis

Infrastructure

Docker Compose

GitHub Actions (CI)

Nominatim

Core Capabilities



Smart Ingestion

Bulk upload free-text statements via CSV/JSON.



Advanced NLP

Extracts entities, events, time, and locations.



Unified Dashboard

Visualizes data across interactive timelines and maps.



100% Traceability

Every flagged contradiction links directly to the source text.

Target Audience & Real-World Value



Investigative Journalists

Cross-reference multiple sources rapidly for complex stories.



Legal Reviewers & Paralegals

Automate the preparation of case statement comparisons.



Insurance Claim Teams

Reconcile accident accounts efficiently.

The Value: Shifts human effort from manual data-entry to high-level interpretation.



Thank You.

Team 66

Silent Witness Capstone Project

[Q&A / Contact]